Science practice booklet: Grade 3 expectations

Grade 3 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 3 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: 3-LS1-1, 3-LS2-1, 3-LS3-1, 3-LS3-2, 3-LS4-1, 3-LS4-2, 3-LS4-3, 3-LS4-4

A worked example

Maryam sees a baby Arabian oryx standing next to its mother. Both have a pale coat and long, straight horns. Is the pale coat an inherited trait or a learned one? Explain.

1. An inherited trait is passed from parents to their young, like the colour of a coat.
2. The baby oryx did not have to learn to have a pale coat. It was born with it.
3. So the pale coat is an inherited trait.

Inherited: the baby is born with the coat its parents have.

Questions

1. Name the stages in the life cycle of a falcon, in order.

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2. A caterpillar grows and then changes into a butterfly. Which stage comes between the caterpillar and the adult butterfly?

   A. Egg
   B. Chrysalis (pupa)
   C. Nymph
   D. Hatchling

3. Put the life of a date palm in order: adult palm with fruit, seed, young palm, seedling.

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4. Arabian oryx live in herds. Which is a way that living in a herd helps an oryx?

   A. A herd keeps the desert cooler
   B. A herd means the oryx never need to drink
   C. Each oryx can eat more food than it could alone
   D. Many animals can watch for danger and protect the young

5. Which of these is a trait that a young camel inherits from its parents?
   A. A hump that stores fat
   B. Kneeling when its trainer says a word
   C. Walking calmly in a line of camels
   D. Knowing the way to a well

6. A falcon is trained to return to its owner's glove. Is this trait inherited or learned?

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7. Two seeds from the same date palm are planted. One gets plenty of water and sunlight, the other gets very little. Which grows taller, and what does this show?

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8. Hamdan finds the fossil of a sea shell in a rock in the UAE desert, far from the sea. What might this tell us about that place long ago?

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9. Why are a camel's wide, padded feet useful on desert sand?
   A. They store water for long journeys
   B. They make the sand cooler under its feet
   C. They spread its weight so it does not sink into soft sand
   D. They help it run faster than any other animal

10. Mangrove trees (القرم) grow in salty mud on the coast. Their roots stick up out of the mud. Why does this help the tree?
    A. The roots let the tree float on the water
    B. The roots take in air, because the mud holds very little
    C. The roots make sea water fresh for people
    D. The roots keep the tree cool

11. A small pond in a wadi dries up in a very hot summer. What might happen to the animals that lived there?

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12. Explain why the pale sandy coat of an Arabian oryx helps it survive in the desert.

Section 2: Earth and space

Standards practised: 3-ESS2-1, 3-ESS2-2, 3-ESS3-1

A worked example

The forecast for Tuesday in Dubai says 38 degrees and sunny, with no rain. Is that weather or climate? Explain.

1. Weather is what the air is like in one place on one day or over a short time.
2. Climate is the pattern of weather over many years.
3. A forecast for one day is about weather.

Weather: it describes one day.

Questions

1. What is the difference between weather and climate?

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2. Which sentence describes climate?

   A. A sandstorm hit Sharjah yesterday afternoon

   B. It is 41 degrees in Abu Dhabi today

   C. Summers in the UAE are usually hot and dry

   D. Rain is forecast for Fujairah tomorrow

3. Which tool measures how much rain has fallen?

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4. Which tool measures how hot the air is?

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5. Rainfall in millimetres from Monday to Friday was: 0, 0, 5, 12, 0. How much rain fell in the five days altogether?

   A. 17 mm

   B. 12 mm

   C. 5 mm

   D. 29 mm

6. Using the same rainfall data, on which day did it rain the most?

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7. The usual daytime temperature in one city is 22 degrees in January and 41 degrees in July. Which month is hotter, and which season is July in?

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8. Which is the best description of the climate of most of the UAE?

   A. Cold winters with heavy snow

   B. Rainy weather all year

   C. Cool and wet summers

   D. Hot summers, mild winters and very little rain

9. A sandstorm is coming while the family is at home. What is the safest thing to do?

   A. Open the windows to let the dust out

   B. Stay indoors and keep the windows and doors closed

   C. Go outside to watch the storm

   D. Play in the garden with a scarf on

10. After very heavy rain, water rushes down a dry wadi. Why should people stay away from it?

    A. Wadis are always full of animals

    B. The water is too salty to touch

    C. Fast water can carry away people and cars, even if it looks shallow

    D. Rain makes the sand too hot to walk on

11. Khalid's family visits a solar park in July, when the weather can be dangerously hot. Give two ways to stay safe from the heat.

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12. Each August for the last five years the top temperature in Abu Dhabi was above 40 degrees. What do you expect for next August? Explain.

Section 3: Matter, forces and energy

Standards practised: 3-PS2-1, 3-PS2-2, 3-PS2-3, 3-PS2-4

A worked example

Salem and Khalid pull a rope in opposite directions in a tug of war. They pull with the same strength. What happens to the rope?

1. Each boy pulls one way with a force.
2. The two forces are the same size and in opposite directions, so they are balanced.
3. Balanced forces do not change the motion of the rope, so it stays still.

The rope does not move.

Questions

1. What do we call two forces that are the same size and push or pull in opposite directions on an object?

   _____

2. In a tug of war, Team A pulls much harder than Team B. What happens to the rope?

   A. It moves towards Team A

   B. It stays still

   C. It moves towards Team B

   D. It spins round and round

3. A book rests on a table and does not move. Which two forces act on it?

   _____

4. Layla kicks a ball that is lying still. What does the kick do to the ball's motion?

   _____

5. Why does a ball that you drop fall to the ground?

   A. The ball runs out of energy

   B. Gravity, a pull from the Earth, pulls it down

   C. The air pushes it down hard

   D. Heavy things want to rest on the ground

6. A swing moves forward and back. One full swing takes 2 seconds. About how many full swings will it make in 10 seconds?

   A. 8

   B. 5

   C. 20

   D. 12

7. Omar notices that a toy car rolled down a ramp stops at about the same place each time he tries. How can he use this pattern?

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8. Hind has four things on the table. Which one will a magnet pick up?

   A. A plastic spoon

   B. A wooden pencil

   C. An aluminium can

   D. A steel paperclip

9. Can a magnet pull a paperclip without touching it? Explain.

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10. What happens when you bring the north ends of two magnets close together? What if you bring a north end and a south end together?

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11. Hind wants a cupboard door to stay shut without a lock. She fixes one magnet on the door and one on the cupboard. How should the magnets face each other?

    A. Two north ends facing each other

    B. Two south ends facing each other

    C. A north end facing a south end

    D. Only one magnet is needed, so no facing is needed

12. Two magnets keep pushing each other away when Noura tries to join them. What can she do to make them stick together? Explain.

Answer guide for parents

Section 1: Living things

1. Egg, then chick, then young falcon, then adult falcon. Each stage grows bigger until the adult can lay eggs.
2. B. Chrysalis (pupa) The caterpillar becomes a chrysalis (pupa) before it is a butterfly. If your child chose C (Nymph), they mixed it up with insects like grasshoppers, which have no chrysalis.
3. Seed, then seedling, then young palm, then adult palm with fruit. The fruit only grows on a fully grown palm, so it comes last.
4. D. Many animals can watch for danger and protect the young. In a herd, many eyes spot danger and adults protect the young. If your child chose B, they think a herd changes what the body needs, but oryx still need water.
5. A. A hump that stores fat. A hump is a body part the camel is born with, so it is inherited. If your child chose B (kneeling when told), they mixed up a trick the camel learns with a trait it is born with.
6. Learned. The falcon was not born doing it, and its trainer teaches it with practice and food.
7. The seed with plenty of water and sunlight grows taller. Both seeds are the same kind, so the difference is the environment.
8. The place was once covered by the sea. A sea shell fossil only forms where a shell lay in sea water long ago.
9. C. They spread its weight so it does not sink into soft sand. Wide feet spread the weight over a bigger area, so the camel does not sink. If your child chose B, they think the feet change the sand, but they only change how the weight is shared.
10. B. The roots take in air, because the mud holds very little. The mud holds very little air, so the roots that stick up take air in. If your child chose A, they think roots help the tree float, but roots are for taking in air.
11. Some animals move to find water elsewhere, and some may not survive. A habitat gives animals water and food, so when it changes they are affected.
12. Pale colours reflect sunlight, so the oryx stays cooler. The coat also matches the sand, so it is harder for others to spot.

Section 2: Earth and space

1. Weather is one day's air, like hot or rainy today. Climate is the usual pattern over many years. Ask: is it about today or about every year?
2. C. Summers in the UAE are usually hot and dry. Usually hot and dry describes a pattern over many years. If your child chose B (41 degrees today), they picked a single day, which is weather.
3. A rain gauge. It is a container that collects the rain, and the water level shows how much fell.
4. A thermometer. It has a liquid or a screen that shows how hot the air is in degrees.
5. A. 17 mm. Only Wednesday and Thursday had rain: 5 + 12 = 17 mm. If your child chose B (12 mm), they picked the biggest day instead of adding.
6. Thursday. The numbers are 0, 0, 5, 12, 0, and 12 is the biggest, which is the fourth day.
7. 41 is bigger than 22, so July is hotter. July is in the middle of the year, which is summer.

8. D. Hot summers, mild winters and very little rain. Hot summers, mild winters and little rain is the pattern in the UAE. If your child chose A, they described a place with snow, which the UAE does not have.
9. B. Stay indoors and keep the windows and doors closed. Closed windows and doors keep the dust out, so everyone stays safe. If your child chose C, they were curious, but a sandstorm can hurt eyes and lungs.
10. C. Fast water can carry away people and cars, even if it looks shallow. Fast water is strong and can carry people and cars away. If your child chose D, they worried about hot sand, but the danger is the moving water.
11. Drink plenty of water and stay in the shade at midday. Light clothes and a hat also keep the body cooler.
12. Probably above 40 degrees again. Five Augusts in a row were hot, so the pattern helps us predict the next one.

Section 3: Matter, forces and energy

1. Balanced forces. They are equal in size and opposite in direction, so they cancel out.
2. A. It moves towards Team A. Team A pulls harder, so the forces are unbalanced and the rope moves towards A. If your child chose B (stays still), they forgot that the pulls are not equal.
3. Gravity pulls the book down and the table pushes it up. The two forces are equal, so the book stays still.
4. The kick is a push, so the ball starts to move. A force can start, stop or change the way something moves.
5. B. Gravity, a pull from the Earth, pulls it down. Gravity is the Earth's pull, and it pulls the ball down. If your child chose A (it runs out of energy), they think the ball tires, but a force makes it fall.
6. B. 5. 10 seconds divided by 2 seconds gives 5 swings. If your child chose C (20), they multiplied 10 × 2 instead of dividing.
7. Omar can predict that the car will stop near the same place again. A pattern that repeats lets us guess what happens next.
8. D. A steel paperclip. Steel contains iron, and a magnet attracts iron. If your child chose C (aluminium can), they think every metal is magnetic, but aluminium is not.
9. Yes. The magnet's force reaches across a small gap, so the paperclip jumps to it before they touch.
10. Two north ends push apart, and a north end and a south end pull together. Same ends repel, different ends attract.
11. C. A north end facing a south end. Different ends attract, so north facing south holds the door shut. If your child chose A, two north ends push apart and the door would not stay closed.
12. Turn one magnet around. Same ends push apart, so turning one puts different ends together and they attract.